

MARISSA ESTEBAN

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TECHNICAL SKILLS & PROFICIENCIES

Python, Matlab, Java, C++, HTML, R, Hardware Design, SQLAlchemy, Pytest, GitHub, Linux, Adobe Software, Figma

TECHNICAL WORK EXPERIENCE AND PROJECTS

Post Baccalaureate Research Fellow, Columbia University INTEGRITY Program

October 2025 - Present

Tools: Python, **PyTorch**

- Developing a deep learning framework for volumetric reconstruction of tissue sections by aligning serial ST and H&E sections in a multimodal latent space
- Built reproducible Python workflows using Scanpy and scVelo to generate high-resolution trajectory maps and identify high-velocity driver genes for single cell and ST datasets

Research Assistant 2, OHSU Neurosurgery

June 2025 - August 2025

Tools: **Matlab**, Ripple Trellis

- Extended open source tools for preprocessing and analyzing sEEG and ECoG recordings to produce consistent, artifact reduced signals for multiple research projects
- Designed hardware mapping workflows to translate native Ripple Neuro channel identifiers into clinical contact names
- Collaborated with researchers to design experimental pipelines which identify and compare highly active regions of the brain during activities that activate humans visual motion system.

Bioinformatics Research Intern, Karolinska Institutet

June 2023 - Aug 2023

Tools: Python, R, Unix, Nextflow (Sarek), Integrative Genomics Viewer

- Utilized a **Nextflow** analysis pipeline in **Linux** and produced the germline and somatic variants, enabling targeted variant detection and certain mutation hotspots.
- Automated data workflows using **Python** and **R** scripts which ensured reproducible **data processing** and **analysis**
- In collaboration with lab scientists and bioinformaticians, efficiently extracted and organized mutation data to ensure compatibility with **spatial transcriptomic** data, identifying 16 heavily mutated cancer genes.
- Delivered insights at the National Biomedical Engineering Society conference, showcasing innovations in mutation mapping and their implications for breast cancer research.

Full Stack Web Developer Intern, Spaced Cadet

June 2022 - Aug 2022

Tools: Python, HTML, Flask, SQAlchemy, PyTest, Bootstrap, **Docker**, Git

- Handled end-to-end development of new features, including a student dashboard and review session which improves retention of class content.
- Ensured robustness and reliability by preparing, documenting, and executing test criteria, test cases, and **test scripts** using **PyTest**.
- Utilized a full stack toolkit, including Flask for server-side operations, **SQLAlchemy** for database management, and Bootstrap for responsive front-end design.

Computer Science Tutor and Lab Assistant, USD School of Engineering

Sep 2021 - May 2024

- Partnered with faculty in over 10 classes to help students acquire confidence in problem-solving and designing programs.
- Assisted students in understanding programming concepts such as algorithm complexities, data structures, and using object-oriented design. Provided new ways of visualizing CS through diagramming and new conceptual frameworks.

WORK EXPERIENCE

Communication Assistant, Advancing Students Forward

Aug 2022 - May 2023

- Established the brand presence of ASF as a binational nonprofit by engaging with supporters and stakeholders through social media
- Analyzed the company's digital marketing plan and social media strategy.

Web Editor & Art Director, USD Student Media

June 2021 - May 2022

- Managed the digital content strategy for multiple platforms, enhancing user engagement through data-driven insights and interactive web elements for the USD Vista, USD TV and USD Radio each week
- Led the design and graphic content creation of the USD Vista each production week to produce 24 12-page papers

EDUCATION

University of San Diego, *Magna Cum Laude*

B.S Computer Science; Minors Art History and Mathematics

ACTIVITIES AND LEADERSHIP

USD Jiu Jitsu Club, *President*

USD Club Swim, *National Qualifier*

Steam Academy, *Program Leader*